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(54) **SYSTEMS AND METHODS FOR ADAPTIVELY CONTROLLING FILAMENT CURRENT IN AN X-RAY TUBE**

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(58) **Field of Classification Search**
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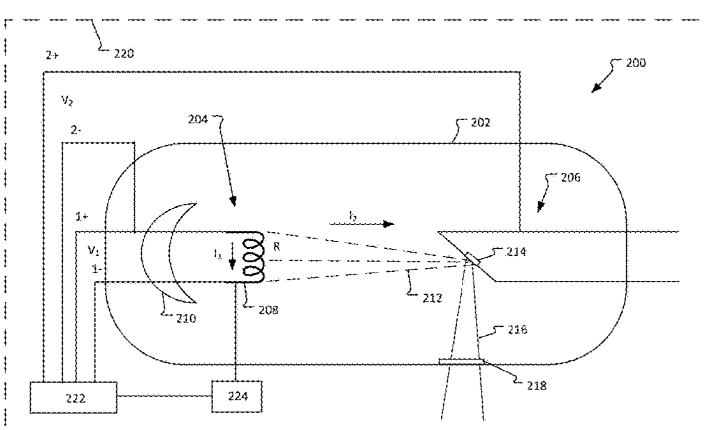
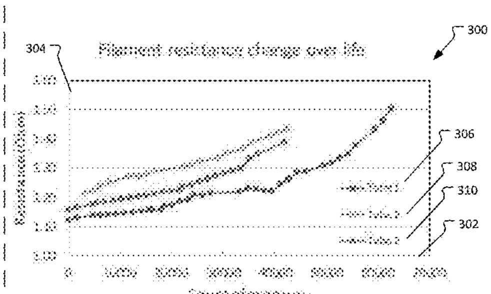
(57) **ABSTRACT**

Systems and methods of adaptively controlling filament current in an x-ray tube of an imaging system include the x-ray tube having a filament being calibrated. Calibration data from the calibration of the x-ray tube is stored at the imaging system, the calibration data including a filament current value that determines a tube current value for a tube voltage value at a plurality of stations. A resistance value of the filament over a period of time is monitored. A change in the resistance value of the filament over the period of time is determined, and the filament current value of at least one of the plurality of stations is adjusted based on the changed resistance value.

19 Claims, 7 Drawing Sheets





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2009/0177495	A1	7/2009	Abousy
2009/0213987	A1	8/2009	Stein et al.
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2009/0281867	A1	11/2009	Sievenpiper
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2010/0020938	A1	1/2010	Koch
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2010/0054400	A1	3/2010	Ren
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2014/0044230	A1	2/2014	Stein et al.
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2019/0200942 A1 7/2019 DeFreitas
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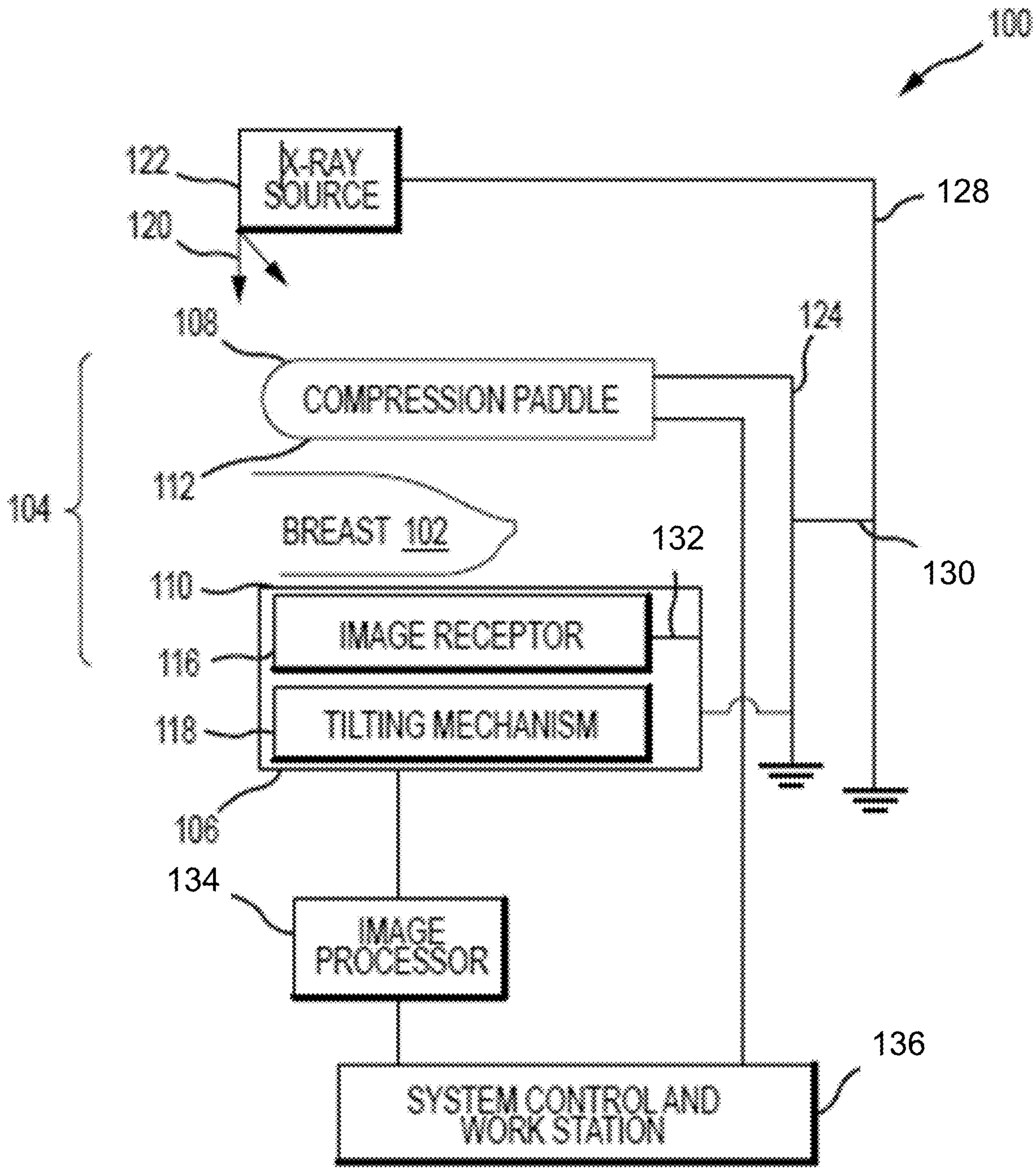

100
122
X-RAY SOURCE
120
128
108
124
COMPRESSION PADDLE
112
104
BREAST 102
130
132
110
IMAGE RECEPTOR
116
118
TILTING MECHANISM
106
134
IMAGE PROCESSOR
136
SYSTEM CONTROL AND WORK STATION

FIG. 1

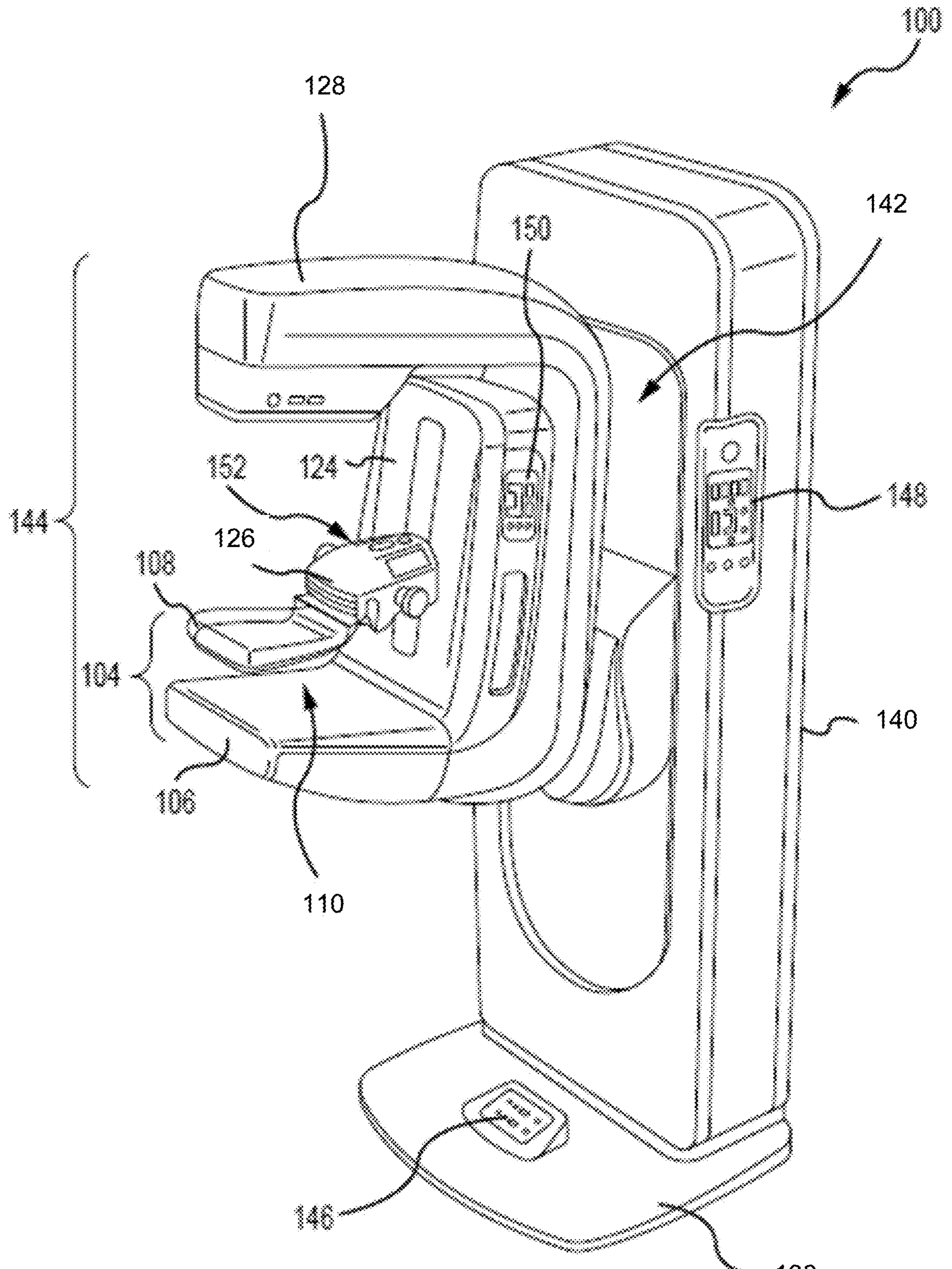

FIG. 2

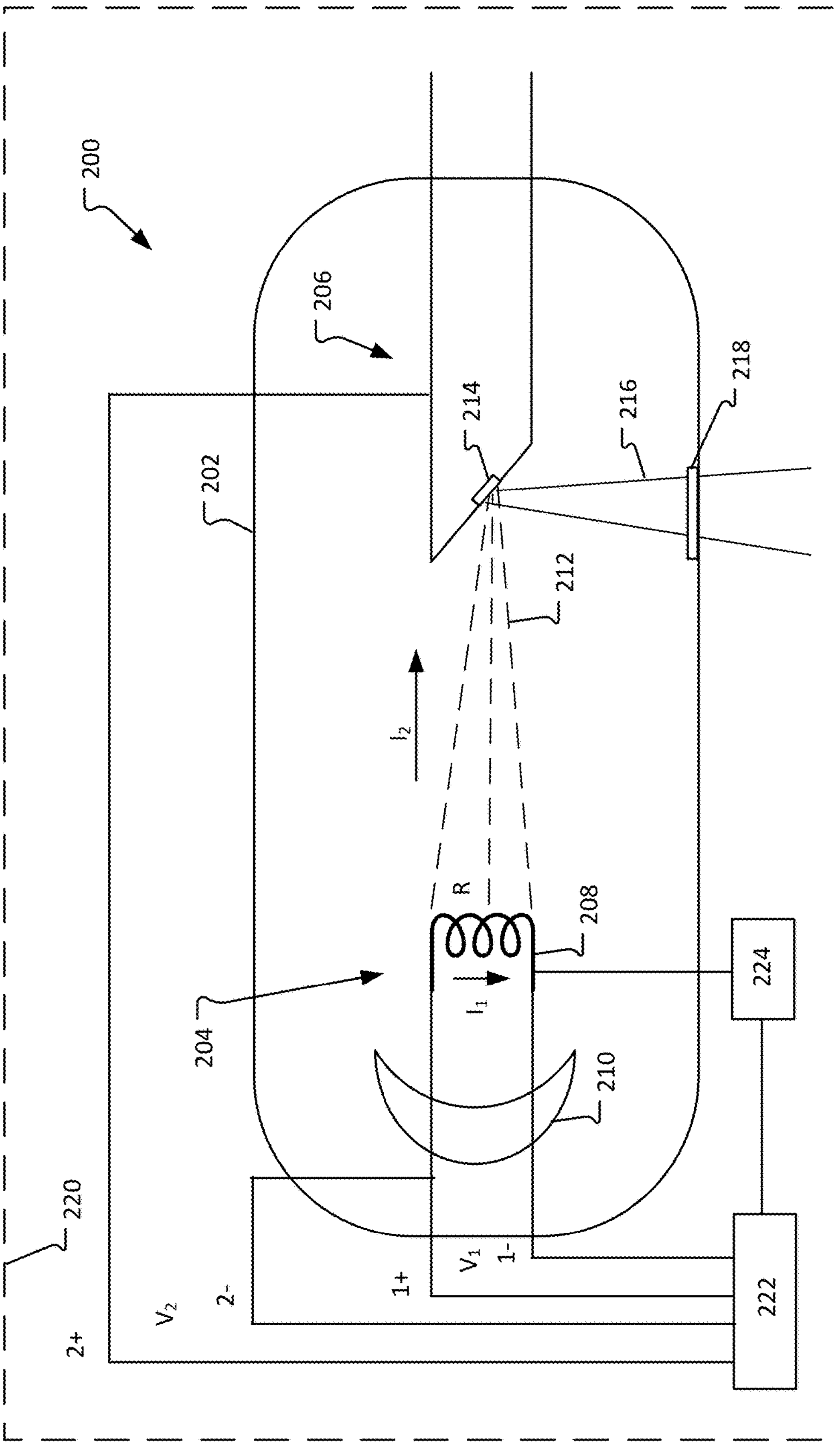
200
206
202
214
216
218
212
I_2
R
208
224
204
I_1
210
220
V_1
1-
1+
2-
V_2
2+
222

FIG. 3

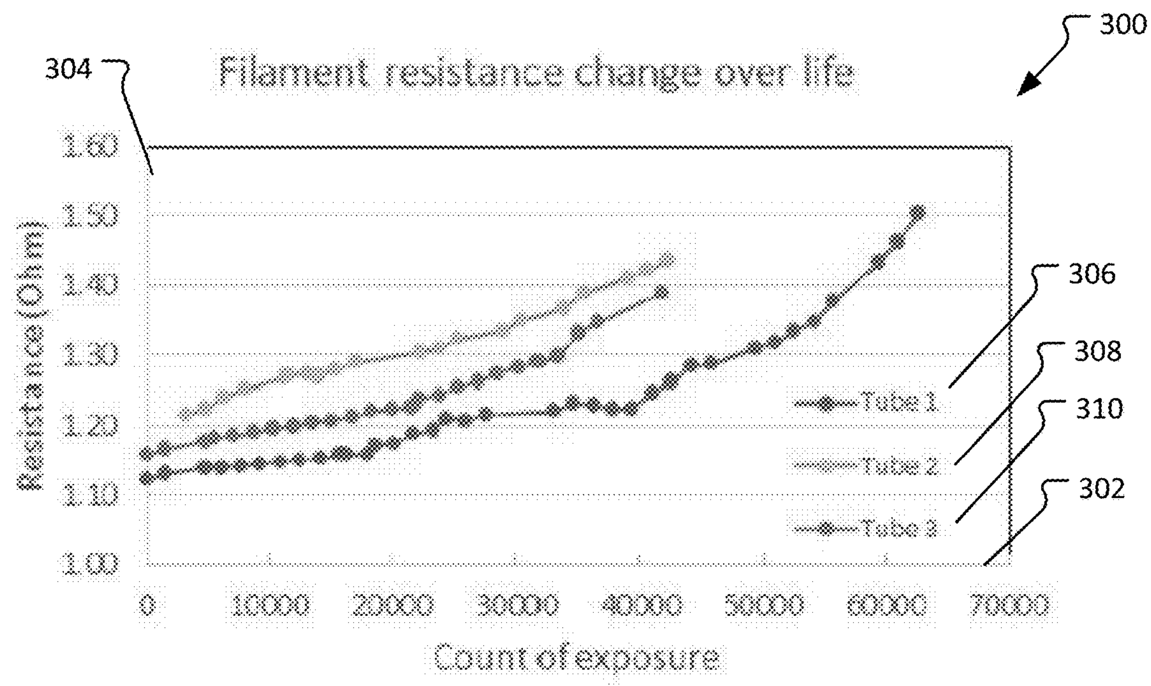
300
Filament resistance change over life
304
Resistance (Ohm)
1.60
1.50
1.40
1.30
1.20
1.10
1.00
0
10000
20000
30000
40000
50000
60000
70000
Count of exposure
Tube 1
Tube 2
Tube 3
306
308
310
302

FIG. 4

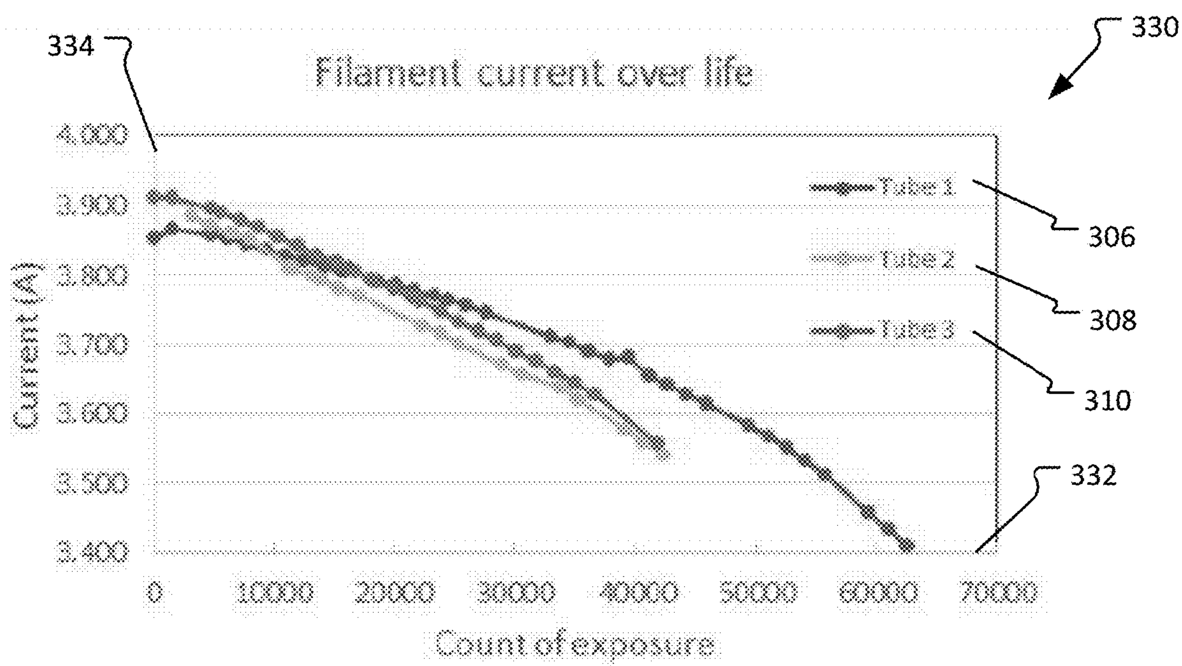
330
Filament current over life
334
Current (A)
4.000
3.900
3.800
3.700
3.600
3.500
3.400
0
10000
20000
30000
40000
50000
60000
70000
Count of exposure
Tube 1
Tube 2
Tube 3
306
308
310
332

FIG. 5

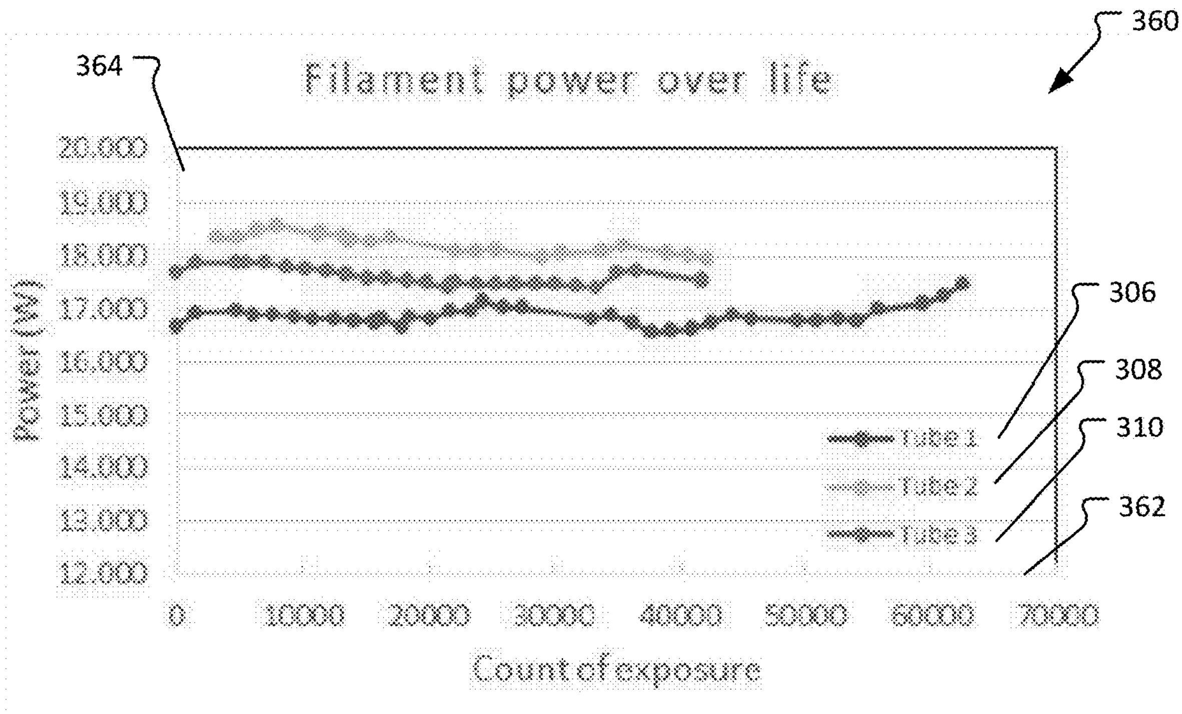
360
364
Filament power over life
20.000
19.000
18.000
17.000
16.000
15.000
14.000
13.000
12.000
Power (W)
0
10000
20000
30000
40000
50000
60000
70000
Count of exposure
Tube 1
Tube 2
Tube 3
306
308
310
362

FIG. 6

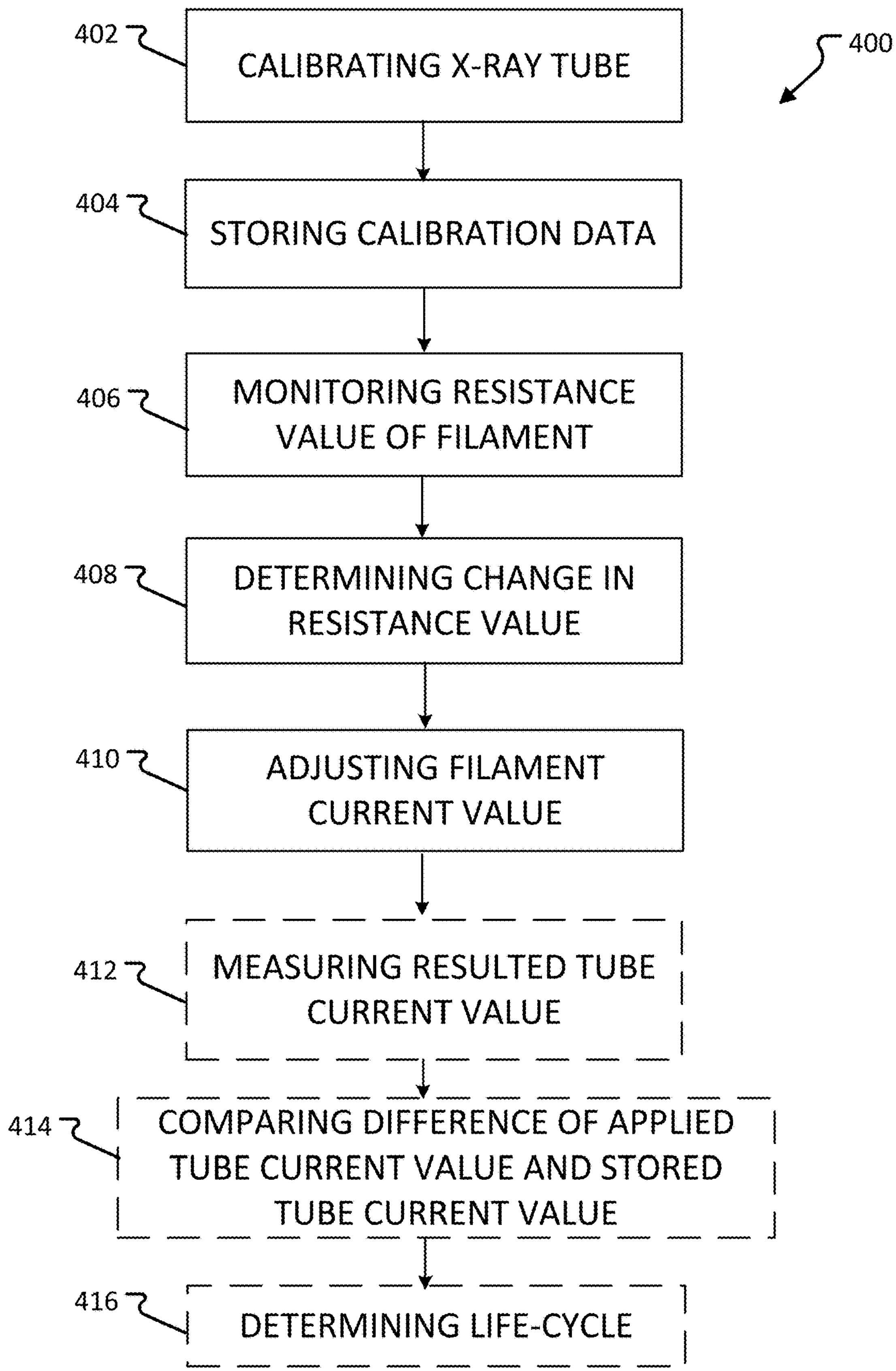
400
402
CALIBRATING X-RAY TUBE
404
STORING CALIBRATION DATA
406
MONITORING RESISTANCE VALUE OF FILAMENT
408
DETERMINING CHANGE IN RESISTANCE VALUE
410
ADJUSTING FILAMENT CURRENT VALUE
412
MEASURING RESULTED TUBE CURRENT VALUE
414
COMPARING DIFFERENCE OF APPLIED TUBE CURRENT VALUE AND STORED TUBE CURRENT VALUE
416
DETERMINING LIFE-CYCLE

FIG. 7

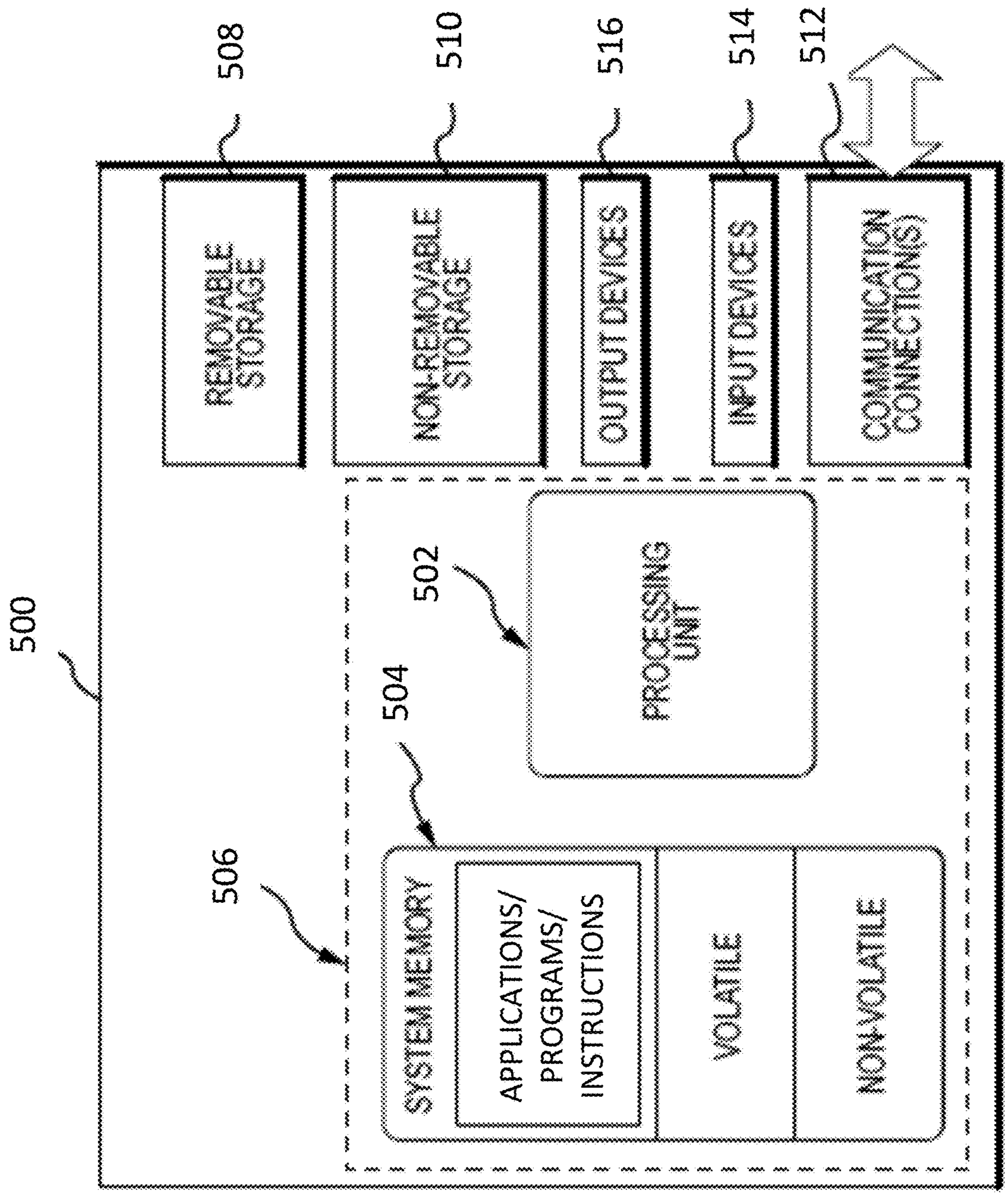
500
502
504
506
508
510
512
514
516
REMOVABLE STORAGE
NON-REMOVABLE STORAGE
OUTPUT DEVICES
INPUT DEVICES
COMMUNICATION CONNECTION(S)
PROCESSING UNIT
SYSTEM MEMORY
APPLICATIONS/ PROGRAMS/ INSTRUCTIONS
VOLATILE
NON-VOLATILE

FIG. 8

SYSTEMS AND METHODS FOR ADAPTIVELY CONTROLLING FILAMENT CURRENT IN AN X-RAY TUBE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to and the benefit of U.S. Provisional Application No. 63/307,311, filed Feb. 7, 2022, which is hereby incorporated by reference herein in its entirety.

BACKGROUND

Imaging based on the use of x-rays is commonplace in medical imaging technology, such as, but not limited to, mammography or tomosynthesis systems. The x-rays used in such imaging technology are often generated through the use of an x-ray tube. Inside the x-ray tube there is an anode and a cathode, and within the cathode there is a filament that emits electrons. The electrons are accelerated by an electrical field generated by applying a high voltage potential across the anode and cathode terminals. As the filament wears with use, the filament material evaporates under high temperature. This evaporation results in a thinning of the material and an increase in filament resistance. When the resistance of the filament increases, power and temperature at the filament are increased, thereby resulting in an increased radiation output.

SUMMARY

In one aspect, the technology relates to a method of adaptively controlling filament current in an x-ray tube of an imaging system, the method including: calibrating the x-ray tube having a filament; storing calibration data from the calibration of the x-ray tube at the imaging system, wherein the calibration data includes a filament current value that determines a tube current value for a tube voltage value at a plurality of stations; monitoring a resistance value of the filament over a period of time; determining a change in the resistance value of the filament over the period of time; and adjusting the filament current value of at least one of the plurality of stations based on the changed resistance value.

In an example, monitoring the resistance value of the filament over the period of time includes measuring the resistance value after a predetermined number of sequence exposures and storing each measured resistance value. In another example, measuring the resistance value after the predetermined number of sequence exposures includes applying a constant filament current value to the filament after each of the predetermined number of sequence exposures. In yet another example, the method further includes waiting a predetermined time period between an end of the predetermined number of sequence exposures and prior to applying the constant filament current value to the filament for resistance value measurement. In still another example, the filament current value of each of the plurality of station is updated based on the changed resistance value. In an example, determining the change in the resistance value includes comparing a difference between resistance values over the period of time to a predetermined benchmark.

In another example, the method further includes measuring a resulted tube current value from the x-ray tube at a sequence exposure having a stored tube current value and tube voltage value for a station of the plurality of stations; and comparing a difference between the measured resulted tube current value and the stored tube current value for the respective station. In yet another example, based on the difference between the measured resulted tube current value and the stored tube current value, the period of time is at least partially defined. In still another example, measuring resulted tube current value is performed pre-exposure or post-exposure of the sequence exposure. In an example, the method further includes determining a life-cycle period of the filament based at least partially on the monitored resistance value of the filament.

In another aspect, the technology relates to an imaging system includes: an x-ray tube having a filament; a current control circuit coupled to the x-ray tube and configured to channel current through the filament; at least one processor communicatively coupled to the current control circuit; and memory communicatively coupled to the at least one processor, the memory including computer executable instructions that, when executed by the at least one processor, performs a method including: calibrating the x-ray tube; storing calibration data from the calibration of the x-ray tube at the imaging system, wherein the calibration data includes a filament current value that determines a tube current value for a tube voltage value at a plurality of stations; monitoring a resistance value of the filament over a period of time; determining a change in the resistance value of the filament over the period of time; and adjusting the filament current value of at least one of the plurality of stations based on the changed resistance value.

In an example, monitoring the resistance value of the filament over the period of time includes measuring the resistance value after a predetermined number of sequence exposures and storing each measured resistance value. In another example, measuring the resistance value after the predetermined number of sequence exposures includes applying a constant filament current value to the filament after each of the predetermined number of sequence exposures. In yet another example, the method further includes waiting a predetermined time period between an end of the predetermined number of sequence exposures and prior to applying the constant filament current value to the filament for resistance value measurement. In still another example, the filament current value of each of the plurality of station is updated based on the changed resistance value. In an example, determining the change in the resistance value includes comparing a difference between resistance values over the period of time to a predetermined benchmark.

In another example, the method further includes: measuring a resulted tube current value from the x-ray tube at a sequence exposure having a stored tube current value and tube voltage value for a station of the plurality of stations; and comparing a difference between the measured resulted tube current value and the stored tube current value for the respective station. In yet another example, based on the difference between the measured resulted tube current value and the stored tube current value, the period of time is at least partially defined. In still another example, measuring resulted tube current value is performed pre-exposure or post-exposure of the sequence exposure. In an example, the method further includes determining a life-cycle period of the filament based at least partially on the monitored resistance value of the filament.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects,

features, and/or advantages of examples will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic view of an exemplary imaging system.

FIG. **2** is a perspective view of the imaging system of FIG. **1**.

FIG. **3** is a schematic view of an x-ray tube.

FIG. **4** is a graph depicting filament resistance change over a life of an x-ray tube.

FIG. **5** is a graph depicting filament current change over the life of an x-ray tube.

FIG. **6** is a graph depicting filament power change over the life of an x-ray tube.

FIG. **7** depicts a flowchart illustrating a method of adaptively controlling filament current in an x-ray tube.

FIG. **8** depicts an example of a suitable operating environment for use with the present examples.

DETAILED DESCRIPTION

As discussed above, x-ray tubes in medical imaging systems have limited lifetimes. The limited lifetime of x-ray tubes is often due to the high heat and high voltages that are generally required for the operation of the x-ray tube. The high heat and voltages cause the components of the x-ray tube to break down, and in some components, change performance characteristics. When the x-ray tube changes performance characteristics, the x-ray tube needs to be recalibrated, and in some instances replaced. Recalibration and/or replacement costs of the x-ray tube are often significant, and according, improvements to the x-ray tube are desired.

Based on analysis of x-ray tube filaments, as the filament degrades with use, the filament material evaporates and reduces its thickness. This results in the resistive properties of the filament changing and increasing. As the resistance of the filament increases, the power and the temperature generated at the filament increases during operation, causing an increase in radiation output from the x-ray tube.

The present technology relates to a feedback mechanism to adjust the filament current needed to maintain the required or desired x-ray radiation at each tube output current and tube voltage station as the filament resistance changes over time. For example, after calibration of the x-ray tube, the filament power (e.g., based on the filament current generated) for each tube output current and tube voltage station is calculated and saved by the control system. Over the operational life of the x-ray tube, the filament resistance is monitored. When the filament resistance changes by a predetermined amount, the filament current may be adjusted appropriately by the control system to get the filament power the same as the stored value for each tube output current and tube voltage station. As such, the x-ray tube can deliver the desired x-ray radiation even with the filament changing resistive characteristics. Additionally, recalibration procedures will be reduced as the x-ray tube degrades with use.

FIG. **1** is a schematic view of an exemplary imaging system **100**. FIG. **2** is a perspective view of the imaging system **100**. Referring concurrently to FIGS. **1** and **2**, not every element described below is depicted in both figures. The imaging system **100** immobilizes a patient's breast **102** for x-ray imaging (either or both of mammography, tomosynthesis, or other imaging modalities) via a breast compression immobilizer unit **104** that includes a breast support platform **106** and a moveable compression paddle **108**. The breast support platform **106** and the compression paddle **108** each have a compression surface **110** and **112**, respectively, that move towards each other to compress, immobilize, stabilize, or otherwise hold and secure the breast **102** during imaging procedures. In known systems, the compression surface **110**, **112** is exposed so as to directly contact the breast **102**. The platform **106** also houses an image receptor **116** and, optionally, a tilting mechanism **118**, and optionally an anti-scatter grid (not depicted, but disposed above the image receptor **116**). The immobilizer unit **104** is in a path of an imaging beam **120** emanating from x-ray source **122**, such that the beam **120** impinges on the image receptor **116**.

The immobilizer unit **104** is supported on a first support arm **124** and the compression paddle **108** is supported on the first support arm **124** by a paddle mount **126**, which is configured to be raised and lowered along the support arm **124**. The x-ray source **122** is supported on a second support arm, also referred to as a tube head **128**. For mammography, support arms **124** and **128** can rotate as a unit about an axis **130** between different imaging orientations such as CC and MLO, so that the system **100** can take a mammogram projection image at each orientation. In operation, the image receptor **116** remains in place relative to the platform **106** while an image is taken. The immobilizer unit **104** releases the breast **102** for movement of arms **124**, **128** to a different imaging orientation. For tomosynthesis, the support arm **124** stays in place, with the breast **102** immobilized and remaining in place, while at least the second support arm **128** rotates the x-ray source **122** relative to the immobilizer unit **104** and the compressed breast **102** about the axis **130**. The system **100** takes plural tomosynthesis projection images of the breast **102** at respective angles of the beam **120** relative to the breast **102**.

Concurrently and optionally, the image receptor **116** may be tilted relative to the breast support platform **106** and in sync with the rotation of the second support arm **128**. The tilting can be through the same angle as the rotation of the x-ray source **122**, but may also be through a different angle selected such that the beam **120** remains substantially in the same position on the image receptor **116** for each of the plural images. The tilting can be about an axis **132**, which can but need not be in the image plane of the image receptor **116**. The tilting mechanism **118** that is coupled to the image receptor **116** can drive the image receptor **116** in a tilting motion. For tomosynthesis imaging and/or CT imaging, the breast support platform **106** can be horizontal or can be at an angle to the horizontal, e.g., at an orientation similar to that for conventional MLO imaging in mammography. The system **100** can be solely a mammography system, a CT system, or solely a tomosynthesis system, other modalities such as ultrasound, or a "combo" system that can perform multiple forms of imaging. An example of a system has been offered by the assignee hereof under the trade name Selenia Dimensions.

Whether operating in a mammography or a tomosynthesis mode, the system **100** images the breast **102** by emitting the x-ray beam **120** from the x-ray source **122**. The x-ray beam **120** passes through the breast **102** where it is detected by the image receptor **116**. The image receptor **116** may include a plurality of pixels that detect the intensity of the x-ray beam **120** at a plurality of locations after the x-ray beam **120** has passed through the breast **102**. The attenuation of the x-ray beam **120** as it passes through the breast **102** changes depending on the structures of the breast **102**. Accordingly, images of the breast **102** may be produced from the detected x-ray beam **120**. For instance, the image receptor **116** produces imaging information in the form of electric signals, and supplies that imaging information to an image processor **134** for processing and generating x-ray images of the breast **102**. A system control and work station unit **136** including software controls the operation of the system **100** and interacts with the operator to receive commands and deliver information including processed-ray images. The system control and work station unit **136** may also include software for controlling the operation of the x-ray source **122**.

The imaging system **100** includes a floor mount or base **138** for supporting the imaging system **100** on a floor. A gantry **140** extends upwards from the floor mount **138** and rotatably supports both the tube head **128** and the support arm **124**. The tube head **128** and support arm **124** are configured to rotate discretely from each other and may also be raised and lowered along a face **142** of the gantry **140** so as to accommodate patients of different heights. The x-ray source **122** is disposed within the tube head **128**. Together, the tube head **128** and support arm **124** may be referred to as a C-arm **144**. A number of interfaces and display screens are disposed on the imaging system **100**. These include a foot display screen **146**, a gantry interface **148**, a support arm interface **150**, and a compression arm interface **152**. In general the various interfaces **146**, **148**, **150**, and **152** may include one or more tactile buttons, knobs, switches, as well as one or more display screens, including capacitive touch screens with graphic user interfaces (GUIs) so as to enable user interaction with and control of the imaging system **100**.

FIG. **3** is a schematic view of an x-ray tube **200**. The x-ray tube **200** may be included as at least part of the x-ray source **122** discussed above. The x-ray tube **200** includes a tube body **202** housing a cathode **204** and an anode **206**. The cathode **204** includes a filament **208** and, in some examples, a focusing cup **210**. The filament **208** can be placed adjacent the focusing cup **210** and between the focusing cup **210** and the anode **206**. The filament **208** may be formed from a material with a high melting point, such as tungsten. A voltage or signal V_1 may be applied across the filament **208** via wires connected to each end of the filament **208**, indicated by the 1+ for the positive connection to the filament **208** and the 1− for the negative connection to the filament **208**. When the signal or voltage is applied across the filament **208**, a current I_1 flows through the filament **208** which heats the filament **208** and causes electrons to be emitted from the filament **208**. Due to a high voltage potential V_2 applied between the cathode **204** and the anode **206** as indicated by the 2+ for the positive connection to the anode **206** and the 2− for the negative connection to the cathode **204**, the electrons emitted from the filament **208** are accelerated towards the anode **206**. The accelerated electrons form an electron beam **212** that travels along an electron beam path between the cathode **204** and the anode **206**. The electron beam **212** impacts the anode **206** at a focal spot **214** and causes the emission of x-rays **216** from the anode **206**. The x-rays **216** exit the x-ray tube body **202** through a tube window **218**. The x-rays **216** that exit from the body **202** form the x-ray beam that is used for imaging, such as the x-ray beam **120** discussed above with reference to FIGS. **1** and **2**.

The area that the electron beam **212** impacts the anode **206** is referred to as the focal spot **214**. This size of the focal spot **214** relates to the resolution required or desired for the imaging process. The location of the focal spot **214** on the anode **206**, as well as, the angle of the anode **206**, have an effect on the direction of the x-rays **216** produced from the anode **206**. The size and location of the focal spot **214** may be controlled or modified by the focusing cup **210**. Additionally, the electron beam **212** is produced by thermionic effect from the filament **208** being heated by the electric current I_1. This current value of the filament **208**, via applied voltage V_1 across the filament **208**, determines a tube output value I_2, typically measured in milliamperes (mA), for a given tube high voltage potential V_2, typically measured in kilovoltage (kV).

In the example, the x-ray tube **200** is included in a high voltage control circuit **220** (also known as a current control circuit) that is configured to control operation of the x-ray tube **200**. The high voltage control circuit **220** may include a micro-computer **222** that stores operational data and processes for the x-ray tube **200** and facilitates operation of the x-ray tube **200**. For example, the high voltage control circuit **220**, via the micro-computer **222** coupled in communication with the x-ray tube **200**, applies voltage V_1 across the filament **208** (e.g. via channeling current through the filament) based on stored calibration data so as to emit x-rays **216** having a required or desired radiation amount (e.g., radiation dose). In x-ray imaging, control of the amount of x-ray radiation for increased image quality is desired. In an aspect, the x-ray dose is proportional to the tube output value I_2 per second (mAs) for a given kV tube voltage potential V_2.

In operation, when a new x-ray tube **200** is installed in an x-ray imaging system, the current applied I_1 (e.g., via voltage V_1 generation) at the filament **208** is calibrated for a plurality of kV tube voltage V_2 and mA tube output I_2 stations. In an aspect, every x-ray tube **200** can have unique performance characteristics, and as such, each x-ray tube may have a slightly different calibration. Additionally, the x-ray tube **200** can also be re-calibrated as required or desired as performance characteristics change. As used herein, a station corresponds to discrete kV tube voltage V_2 and mA tube output I_2 values that produce, and reproduce, the same quantity of radiation during an exposure. In an aspect, power applied at the filament **208** is based at least partially on filament current I_1 and controls the tube output current values. However, as the filament **208** wears and degrades with use, the filament material evaporates under high temperature, which results in a thinning of the filament material and in turn, an increase of filament resistance R. As the resistance R increases in the filament **208**, during operation of the x-ray tube, the power (e.g., via the voltage V_1 applied) generated at the filament **208** increases and the temperature of the filament **208** increases, thereby resulting in an increased radiation output from the same voltage V_1 being applied across the filament **208**.

In some systems, the high voltage control circuit **220** may measure the mA tube output I_2 of the x-ray tube after every exposure sequence (e.g. via an ammeter or the like). As used herein, an exposure sequence is the x-ray emissions during a modality procedure. Accordingly, in some modalities, such as mammography, an exposure sequence may be a single emission and the mA tube output I_2 measurement occurs prior to firing the x-ray. In other modalities, such as tomosynthesis, an exposure sequence may be a plurality of emissions and the mA tube output I_2 measurement occurs after firing the x-rays. This measured actual mA output value I_2 of the exposure sequence may then be compared to the desired mA output value I_2 for the exposure sequence. If the difference between the actual mA output value I_2 and the desired mA output value I_2 exceeds a predetermined threshold, then the control circuit **220** may prompt for a new filament calibration. In an aspect, the predetermined threshold may be based on how much different the actual output is from the desired output while still maintaining accurate and usable x-ray images. In some examples, the predetermined threshold may be between about 35%-20%. In other examples, the predetermined threshold may be about 33%. However, if the difference between the actual mA output value I_2 and the desired mA output value I_2 does not exceed the predetermined threshold, then the control circuit **220** can adjust the voltage V_1 being applied to the filament **208** for the next exposure sequence. This voltage adjustment by the control circuit **220** enables the subsequent measured mA output values I_2 to be closer to the desired mA output values I_2 and increase x-ray imaging performance.

Adjusting the voltage V_1 applied to the filament **208** in the process described above, however, only adjusts the voltage V_1 for the active kV tube voltage V_2 and mA tube output I_2 station relating to the specific the exposure sequence. The other stations (e.g., inactive stations) that are not in use do not gain the benefit of the voltage adjustment algorithm. Furthermore, the mA tube output I_2 measurements only account for the fact that the mA output has changed in the x-ray tube **200** and does not address the underlying tube characteristics that caused the changed mA output, and as such, the accuracy of the adjustments are reduced. Accordingly and as described above, it is the resistive properties R of the filament **208** that change over time and result in the changing mA output values I_2 of the x-ray tube **200**. Thus, and in the examples described herein, the high voltage control circuit **220** is configured to monitor the resistive properties R of the filament **208**. As such, the current I_1 applied to the filament **208** (e.g., via voltage V_1) may be adjusted based on the monitoring of the resistance properties R of the filament **208**. By using the resistive properties R of the filament **208**, each and every kV tube voltage V_2 and mA tube output I_2 station can be adjusted, thereby increasing x-ray imaging performance. Additionally, the accuracy of the adjustments from the x-ray tube calibration baseline values are increased because the adjustments are based on the underlying changing characteristics of the filament **208** that cause the change in mA output values I_2.

In the example, the high voltage control circuit **220** can include a monitoring instrument **224** operatively controlled by the micro-computer **222** to monitor the resistance values R of the filament **208**. In an aspect, the monitoring instrument **224** may be a multimeter or the like that is configured to measure electrical properties of the filament **208**, such as one or more of voltage, resistance, and current. Additionally or alternatively, the monitoring instrument **224** can calculate resistance R of the filament **208** using Ohm's law when the current and voltage applied across the filament **208** are known. The control circuit **220** also may be configured with a clock or other time keeping instrument whereby elapsed time may be monitored and certain operations can be performed at predetermined time sampling periods, intervals, or cycles. Accordingly, when the resistance R of the filament **208** changes by a certain amount over time, an algorithm for the control circuit **220** can adjust the filament current I_1, via the voltage V_1, to get the filament power the same as the stored calibration value for each kV tube voltage V_2 and mA tube output I_2 station. This leads to the x-ray tube **200** being capable of delivering the required or desired mA tube output value I_2 even with changing filament resistance values R.

FIG. **4** is a graph **300** depicting filament resistance change over a life of the x-ray tube **200** (shown in FIG. **3**). As described above, the change in the resistance properties of the filament **208** (shown in FIG. **3**) changes over the operational life of the x-ray tube **200** due to evaporation of filament material under high temperatures. The graph **300** has an x-axis **302** that counts the number of exposures of the x-ray tube **200**, and thus, is a time component for the filament **208**. In the example, the x-ray exposures are at a 33 kV tube voltage and a 200 mA tube current with 60 mAs. The mAs is the product of tube current in mA and time in seconds. A y-axis **304** charts the resistance in Ohms of the filament **208**. As illustrated in the graph **300**, three different x-ray tubes **306**, **308**, **310** are shown to have their filament resistance change during the operational use of the x-ray tubes. Each x-ray tube **306**, **308**, **310** is slightly different because all x-ray tubes are unique due to the design and manufacture of the individual x-ray tubes. However, all of the x-ray tubes **306**, **308**, **310** demonstrate that filament resistance values generally increase over time. This time period of change is over a period ten-thousand plus exposure counts, and thus, filament resistance does not change noticeably after every exposure count.

As used herein, the term "life" or "life-cycle" of the x-ray tube does not necessarily mean the life cycle to a point of physical failure. Rather, "life" or "life cycle" refers to a period of usage after which the x-ray tube no longer performs as required or desired.

FIG. **5** is a graph **330** depicting filament current change over the life of the x-ray tube **200** (shown in FIG. **3**). As discussed in reference to FIG. **4** above, the resistance properties of the filament **208** (shown in FIG. **3**) changes during operational use of the x-ray tube **200**. Accordingly, as the filament resistance changes, the current applied at the filament **208** has to be changed during the operational use of the x-ray tube **200** to get desired tube output. The graph **330** has an x-axis **332** that counts the number of exposures of the x-ray tube **200**, and thus, is a time component for the filament **208**. A y-axis **334** charts the current in amperes of the filament **208**. The graph **330** shows the filament current of the same three x-ray tubes **306**, **308**, **310** as FIG. **4** and at 33 kV tube voltage and 200 mA tube output. The curves illustrate the change in the filament current needed to maintain a constant 200 mA tube output as the filament resistance value increases over time.

Typically, when a new x-ray tube is installed in an imaging system, the filament current is calibrated for the required or desired kV tube voltage and mA tube current stations and these calibration values are stored within the system so that imaging can occur with the correct amount of x-ray radiation. This calibration process also occurs during recalibration of the x-ray tube **200**. However, as shown in the graphs **300**, **330**, the resistive properties of the filament **208** changes and the filament current being applied at the filament **208** has to be reduced in order to deliver the required or desired mA tube output value. This is because as the filament resistance rises, the power at the filament **208** increases (e.g., power $P=I_2 \times R$) and the temperature of the filament increases if filament current is not adjusted, resulting in increased radiation output at the x-ray tube **200**. As described herein, monitoring the change in resistance values of the filament **208** and generating feedback control of the x-ray tube **200** based on the filament resistance allows the filament current to be adjusted to maintain a required or desired x-ray radiation at each kV tube voltage and mA tube output station.

FIG. **6** is a graph **360** depicting filament power change over the life of the x-ray tube **200** (shown in FIG. **3**). The graph **360** has an x-axis **362** that counts the number of exposures of the x-ray tube **200**, and thus, is a time component for the filament **208** (shown in FIG. **3**). A y-axis **364** charts the power in watts of the filament **208**. The graph **360** shows the filament power of the same three x-ray tubes **306**, **308**, **310** as FIGS. **4** and **5**, and at a 33 kV tube voltage and 200 mA tube output. These curves demonstrate that to maintain the same mA tube output over the life of the x-ray tube **200** it is required to maintain the same filament power as right after filament current calibration for each kV tube voltage and mA tube output station. As such, to keep mA tube output within a required or desired value with filament resistance increasing over time, the filament current must be adjusted periodically to operate at a proper filament current for each station.

Referring to FIGS. **4-6**, the graphs illustrate that filament current adjustment over life of the filament is needed to get a required or desired tube current mA. While FIGS. **4-6** are illustrative of a single station, the algorithm described herein can adjust filament current across active and inactive stations within the imaging system.

FIG. **7** depicts a flowchart illustrating a method **400** of adaptively controlling filament current in an x-ray tube. The example methods and operations can be implemented or performed by the systems and devices described herein (e.g., the imaging system **100** and x-ray tube **200** shown in FIGS. **1-3**). The method **400** begins with calibrating (or re-calibrating) the x-ray tube having a filament therein (operation **402**) and storing the calibration data at the imaging system (operation **404**). In an aspect, the calibration data includes a filament current value that determines a tube output value for a tube voltage value at a plurality of stations that is generated from the calibration of the x-ray tube. Each station is defined by the tube output value in mA and the tube voltage value in kV so that a required or desired x-ray radiation amount is generated by the x-ray tube. In some examples, the filament current value may include a filament voltage value that is applied across the filament to generate the current value. The x-ray calibration procedure can be performed by methods that are currently known (e.g., a substitution method) or developed in the future. Because every x-ray tube has slightly different performance characteristics, calibration enables accurate operation of the imaging system as described herein.

The method **400** continues with monitoring a resistance value of the filament over a period of time (operation **406**). As described above, the filament material evaporates under high temperatures and results in a thinning of the filament material, thereby increasing filament resistance over the operational lifetime of the x-ray tube. In an aspect, monitoring the resistance value of the filament may include measuring the resistance value after a predetermined number of sequence exposures and storing each measured resistance value. Based on the stored resistance values a resistance curve can be generated. In an example, the period of time for monitoring the resistance value can be based on the number of exposure counts and for example resistance values can be measured every 100 sequence exposures since it is known that resistance changes in an order many times this predetermined number. In other examples, the resistance value can be measured after every sequence exposure and stored as required or desired. As such, measuring and storing the resistance value is performed periodically.

In the example, measuring the resistance value includes applying a constant and known filament current value to the filament so as to measure the resistance. In an aspect, filament voltage is applied so that the filament receives about 2.5 amps to measure the resistance since voltage, current, and resistance are related via Ohm's law. Because each sequence exposure generates heat at the filament, in some examples, prior to measuring the resistance value, the filament is allowed to cool so as to increase accuracy of the resistance value measurement. In an example, cooling the filament can include waiting a predetermined time period between an end of the sequence exposures and prior to applying the constant filament current value for measuring the resistance value of the filament.

Turning back to the method **400**, once the resistance value of the filament is being monitored (operation **406**), a change in the resistance value of the filament is determined over the period of time (operation **408**). Because the resistive characteristics of the filament change slowly after every exposure, and the tolerance for each exposure does allow for some change to the tube current output, the filament current value does not need to be revised or adjusted for every exposure count. As such, the change in resistance value can be determined by comparing a difference between resistance values over a period of time to a predetermined benchmark. For example, the predetermined benchmark may be a 4% resistance value change over at least 10,000 exposure counts. It should be appreciated that other predetermined benchmark values are also contemplated herein. For example, but not limiting, the resistance value change may be 2%, 5%, 10%, or the like, and the exposure count may be 5,000, 8,000, 12,000, or the like as required or desired.

Once it has been determined that the resistance value of the filament has changed (operation **408**), then the filament current value is adjusted for at least one of the plurality of stations based on the changed resistance value (operation **410**). By adjusting the filament current applied at the filament, the power at the filament during operation can be the same as the calibrated data for the tube current value and the tube voltage value for the station. Accordingly, the x-ray tube can deliver the required or desired tube current value even with an ever changing filament resistance value. In an aspect, the filament current value of each of the plurality of stations is updated based on the changed resistance value. As such, the calibration data can be adjusted based on the current resistive properties of the filament, and less active stations are still configured for a more accurate tube current value output the next time those stations are used for imaging procedures. In an aspect, adjusting the filament current value is based on a power model, whereby the filament power is maintained as right after initial filament calibration.

In the example, once the filament current value is adjusted the process may repeat itself with monitoring the resistance value over a new and reset period of time, determining change in the resistance value, and further adjust the filament current value as required or desired. This feedback algorithm for the resistance value enables the imaging system to operate longer between maintenance operations that calibrate/re-calibrate the x-ray tube.

In aspects, the method **400** may further include measuring a resulted tube current value from the x-ray tube at a sequence exposure having a stored tube current value and tube voltage value for a station of the plurality of stations (operation **412**), and comparing a difference between the measured resulted tube current value and the stored tube current value for the respective station (operation **414**). By measuring the resulted mA tube output current from the x-ray tube during operation and comparing the measured value to the stored desired output, operational data of the x-ray tube may be generated. For example, the accuracy of the feedback algorithm can be verified by this operational data, and if the adjusted filament current value results in less accurate tube current output values, a new filament calibration may be required. In another example, if the difference between the measured tube output current and the desired output is higher than a predetermined threshold, a new filament calibration may be required.

In other examples, based on the difference between the measured resulted tube current value and the stored tube current value, the period of time is at least partially defined. For example, instead of monitoring the resistance value based on the number of exposure counts, the resistance value can be monitored based on the operational results of the tube head, and if the difference between the measured tube current and the stored desired tube current exceeds a predetermined threshold, the change in resistance value can be used to adjust the filament current value as described above. In aspects, the measurement of the resulted tube current value may be performed pre-exposure or post-exposure of the sequence exposure. For example, in tomosynthesis imaging modalities the tube current measurements may be performed post-exposure, while in mammography imaging modalities the tube current measurement may be performed pre-exposure.

The method **400** may also include using the monitored resistance value of the filament to determine a life-cycle period of the filament (operation **416**). For example, because filament resistance increases during the operational life of the filament due to filament material evaporating, an upper limit of filament resistance can be defined that relates to the filament material being too thin and triggering a tube replacement and a new filament calibration for the new x-ray tube.

FIG. **8** illustrates an exemplary suitable operating environment **500** for controlling an x-ray tube. This operating environment may be incorporated directly into the high voltage control circuit disclosed herein, or may be incorporated into a computer system discrete from, but used to control the breast imaging systems described herein. Such computer system may be, for example, the system control and work station **136** depicted in FIG. **1**. This is only one example of a suitable operating environment and is not intended to suggest any limitation as to the scope of use or functionality. Other well-known computing systems, environments, and/or configurations that can be suitable for use include, but are not limited to, imaging systems, personal computers, server computers, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, programmable consumer electronics such as smart phones, network PCs, minicomputers, mainframe computers, tablets, distributed computing environments that include any of the above systems or devices, and the like.

In its most basic configuration, operating environment **500** typically includes at least one processing unit **502** and memory **504**. Depending on the exact configuration and type of computing device, memory **504** (storing, among other things, instructions to read from data storage devices or sensors, or perform other methods disclosed herein) can be volatile (such as RAM), non-volatile (such as ROM, flash memory, etc.), or some combination of the two. This most basic configuration is illustrated in FIG. **8** by dashed line **506**. Further, environment **500** can also include storage devices (removable, **508**, and/or non-removable, **510**) including, but not limited to, magnetic or optical disks or tape. Similarly, environment **500** can also have input device(s) **514** such as touch screens, keyboard, mouse, pen, voice input, etc., and/or output device(s) **516** such as a display, speakers, printer, etc. Also included in the environment can be one or more communication connections **512**, such as LAN, WAN, point to point, Bluetooth, RF, etc. In embodiments, the connections may be operable to facility point-to-point communications, connection-oriented communications, connectionless communications, etc.

Operating environment **500** typically includes at least some form of computer readable media. Computer readable media can be any available media that can be accessed by processing unit **502** or other devices having the operating environment. By way of example, and not limitation, computer readable media can include computer storage media and communication media. Computer storage media includes volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state storage, or any other tangible medium which can be used to store the desired information. Computer storage media does not include communication media.

Communication media embodies computer readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term "modulated data signal" means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of the any of the above should also be included within the scope of computer readable media. A computer-readable device is a hardware device incorporating computer storage media.

The operating environment **500** can be a single computer operating in a networked environment using logical connections to one or more remote computers. The remote computer can be a personal computer, a server, a router, a network PC, a peer device or other common network node, and typically includes many or all of the elements described above as well as others not so mentioned. The logical connections can include any method supported by available communications media. Such networking environments are commonplace in offices, enterprise-wide computer networks, intranets and the Internet.

The embodiments described herein may be employed using software, hardware, or a combination of software and hardware to implement and perform the systems and methods disclosed herein. Although specific devices have been recited throughout the disclosure as performing specific functions, one of skill in the art will appreciate that these devices are provided for illustrative purposes, and other devices may be employed to perform the functionality disclosed herein without departing from the scope of the disclosure. In addition, some aspects of the present disclosure are described above with reference to block diagrams and/or operational illustrations of systems and methods according to aspects of this disclosure. The functions, operations, and/or acts noted in the blocks may occur out of the order that is shown in any respective flowchart. For example, two blocks shown in succession may in fact be executed or performed substantially concurrently or in reverse order, depending on the functionality and implementation involved.

This disclosure describes some examples of the present technology with reference to the accompanying drawings, in which only some of the possible examples were shown. Other aspects may, however, be embodied in many different forms and should not be construed as limited to the examples set forth herein. Rather, these examples were provided so that this disclosure was thorough and complete and fully conveyed the scope of the possible examples to those skilled in the art. Further, as used herein and in the claims, the phrase "at least one of element A, element B, or element C" is intended to convey any of: element A, element B, element C, elements A and B, elements A and C, elements B and C, and elements A, B, and C. Additionally, one having skill in the art will understand the degree to which terms such as "about" or "substantially" convey in light of the measurement techniques utilized herein. To the extend such terms may not be clearly defined or understood by one having skill in the art, the term "about" shall mean plus or minus ten percent.

Although specific examples were described herein, the scope of the technology is not limited to those specific examples. One skilled in the art will recognize other examples or improvements that are within the scope and spirit of the present technology. Therefore, the specific structure, acts, or media are disclosed only as illustrative examples. Examples according to the technology may also combine elements or components of those that are disclosed in general but not expressly exemplified in combination, unless otherwise stated herein. The scope of the technology is defined by the following claims and any equivalents therein.

What is claimed is:

1. A method of adaptively controlling filament current in an x-ray tube of an imaging system, the method comprising:
- calibrating the x-ray tube having a filament;
- storing calibration data from the calibration of the x-ray tube at the imaging system, wherein the calibration data includes a filament current value that determines a tube current value for a tube voltage value at a plurality of exposure values;
- monitoring a resistance value of the filament over a period of time, wherein monitoring the resistance value includes measuring the resistance value of the filament after a predetermined number of sequence exposures and storing each measured resistance value;
- determining a change in the resistance value of the filament over the period of time, wherein the determined changed resistance value is based on a difference between the stored measured resistance values over a number of exposure counts; and
- adjusting the filament current value of at least one of the plurality of exposure values based on the determined changed resistance value.

2. The method of claim **1**, wherein measuring the resistance value after the predetermined number of sequence exposures includes applying a constant filament current value to the filament after each of the predetermined number of sequence exposures.

3. The method of claim **2**, further comprising waiting a predetermined time period between an end of the predetermined number of sequence exposures and prior to applying the constant filament current value to the filament for resistance value measurement.

4. The method of claim **1**, wherein the filament current value of each of the plurality of exposure values is updated based on the changed resistance value.

5. The method of claim **1**, wherein determining the change in the resistance value includes comparing a difference between resistance values over the period of time to a predetermined benchmark.

6. The method of claim **1**, further comprising:
- measuring a resulted tube current value from the x-ray tube at a sequence exposure having a stored tube current value and tube voltage value for an exposure value of the plurality of exposure values; and
- comparing a difference between the measured resulted tube current value and the stored tube current value for the respective exposure value.

7. The method of claim **6**, wherein based on the difference between the measured resulted tube current value and the stored tube current value, the period of time is at least partially defined.

8. The method of claim **6**, wherein measuring resulted tube current value is performed pre-exposure or post-exposure of the sequence exposure.

9. The method of claim **1**, further comprising determining a life-cycle period of the filament based at least partially on the monitored resistance value of the filament.

10. An imaging system comprising:
- an x-ray tube having a filament;
- a current control circuit coupled to the x-ray tube and configured to channel current through the filament;
- at least one processor communicatively coupled to the current control circuit; and
- memory communicatively coupled to the at least one processor, the memory comprising computer executable instructions that, when executed by the at least one processor, performs a method comprising:
  - calibrating the x-ray tube;
  - storing calibration data from the calibration of the x-ray tube at the imaging system, wherein the calibration data includes a filament current value that determines a tube current value for a tube voltage value at a plurality of exposure values;
  - monitoring a resistance value of the filament over a period of time, wherein monitoring the resistance value includes measuring the resistance value of the filament after a predetermined number of sequence exposures and storing each measured resistance value;
  - determining a change in the resistance value of the filament over the period of time, wherein the determined changed resistance value is based on a difference between the stored measured resistance values over a number of exposure counts; and
  - adjusting the filament current value of at least one of the plurality of stations exposure values based on the determined changed resistance value.

11. The imaging system of claim **10**, wherein measuring the resistance value after the predetermined number of sequence exposures includes applying a constant filament current value to the filament after each of the predetermined number of sequence exposures.

12. The imaging system of claim **11**, wherein the method further comprises waiting a predetermined time period between an end of the predetermined number of sequence exposures and prior to applying the constant filament current value to the filament for resistance value measurement.

13. The imaging system of claim **11**, wherein the filament current value of each of the plurality of exposure values is updated based on the changed resistance value.

14. The imaging system of claim **10**, wherein determining the change in the resistance value includes comparing a difference between resistance values over the period of time to a predetermined benchmark.

15. The imaging system of claim **10**, wherein the method further comprises:

measuring a resulted tube current value from the x-ray tube at a sequence exposure having a stored tube current value and tube voltage value for an exposure value of the plurality of exposure values; and

comparing a difference between the measured resulted tube current value and the stored tube current value for the respective exposure value.

16. The imaging system of claim **15**, wherein based on the difference between the measured resulted tube current value and the stored tube current value, the period of time is at least partially defined.

17. The imaging system of claim **15**, wherein measuring resulted tube current value is performed pre-exposure or post-exposure of the sequence exposure.

18. The imaging system of claim **10**, wherein the method further comprises determining a life-cycle period of the filament based at least partially on the monitored resistance value of the filament.

19. A method of adaptively controlling filament current in an x-ray tube of an imaging system, the method comprising:

calibrating the x-ray tube having a filament;

storing calibration data from the calibration of the x-ray tube at the imaging system, wherein the calibration data includes a filament current value that determines a tube current value for a tube voltage value at a plurality of exposure values;

monitoring a resistance value of the filament over a period of time;

determining a change in the resistance value of the filament over the period of time;

adjusting the filament current value of at least one of the plurality of exposure values based on the changed resistance value;

measuring a resulted tube current value from the x-ray tube at a sequence exposure having a stored tube current value and tube voltage value for an exposure value of the plurality of exposure values; and

comparing a difference between the measured resulted tube current value and the stored tube current value for the respective exposure value, wherein based on the difference between the measured resulted tube current value and the stored tube current value, the period of time is at least partially defined.

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